

HW 06

Lecture 06

Due: Wednesday, July 1 at 11:59pm

AI use

Refer to the syllabus for our AI use policy. You will be asked to include any AI prompts used in this assignment at the end.

Starting a new Quarto file

For this assignment you will begin a new Quarto file. To do this, go to RStudio and click File > New File > Quarto Document. A pop-up will appear asking for a title, author, and format. You may leave these as the default options for now. Click OK and a new file will open. Update the YAML with your title, name, and date.

Exercise 1

Determine which probability distribution would be most appropriate for modeling the quantity of interest: Bernoulli, Binomial, Normal, or Poisson. No explanation is required (note, it may be the case that not all assumptions are satisfied for your distribution of choice; that's ok - for here, choose the most appropriate choice).

- (A) The mean proportion of people who are left-handed
- (B) The number of students in a classroom who have diabetes
- (C) How many previous heart attacks a patient has had in the past decade
- (D) The number of flights taken by a person in a year
- (E) The average weight of newborn infants
- (F) Whether a patient develops lung cancer during their life given that they are a smoker

Exercise 2

Blood cholesterol is often used as a marker for cardiovascular disease risk. Suppose the distribution of total cholesterol levels in adults is approximately normal with:

- Mean (μ) = 200 mg/dL
- Standard deviation (σ) = 30 mg/dL

A patient's cholesterol level is measured at **260 mg/dL**.

- Calculate the **z-score** for this patient's cholesterol value. Use R for your computation and show your code.
- Interpret the z-score: how many standard deviations above or below the mean is this value?
- Based on the z-score, would you consider this value to be **unusually high**?
(Hint: Recall that values more than ~2 standard deviations from the mean are often considered unusual.)

Exercise 3

For this exercise and Exercise 4, you will be using the NC bike crash data set from Lab 1. As a reminder, the code chunk below will tell R we are using the tidyverse library, and save the data under the name ncbikecrash.

```
library(tidyverse)
ncbikecrash <- read.csv("https://mhoch422.github.io/BIOS600_SSII/labs/data/bikecrash.csv")
```

What is the mean hour for when a crash occurs in Durham County for each month? Create a table which displays this information, arranged from earliest to latest hour.

Exercise 4

Create a new variable named `time_crash` based on the hour (`crash_hour`) that the crash occurred and **save in a new dataset** called `ncbikecrash_time`. If the crash occurred during from 12:00 am to 6:59 am, define `time_crash` to be "Early morning." If the crash occurred from 7:00 am to 11:59 am, define `time_crash` to be "Morning." From 12:00 pm to 5:59 pm "Afternoon," and from 6:00 pm to 11:59 pm "Evening."

Then, create a table which counts the number of crashes that occurred during these time periods. When do most crashes occur? **Note:** For this question, you should have one section of code for defining the new dataset with the new variable, and a second section of code for creating the table. (They can be in the same code chunk, but they'll be two separate pieces.)

AI Attestation

Was AI used on this assignment? If so, describe your prompts below. If not, simply state "AI was not used on this assignment."

Submission

As you've seen previously, we can **Render** into an .html file that can be opened by any web browser. To export it as a .pdf, open the file in your web browser and then print to or save as a .pdf document. Contact me in Ed Discussion if you need help!

You will submit the PDF documents for labs and homework to Gradescope as part of your final submission.

To submit your assignment:

- Access Gradescope through the menu on the BIOS 600 Canvas site.
- Click on the assignment, and you'll be prompted to submit it.
- Mark the pages associated with each exercise. All of the pages of your lab should be associated with at least one question (i.e., should be "checked").
- Select the first page of your .PDF submission to be associated with the "Formatting" section.

Grading

Component	Points
Ex 1	6
Ex 2	3
Ex 3	3
Ex 4	3
AI Attestation	1
Formatting	3

The "Formatting" grade is to assess the document format. This includes having a neatly organized document (no excessive output, warnings/messages when loading packages and/or data) with readable code and your name and the date updated in the YAML.